

Word sense disambiguation: process of determining the intended meaning, or sense, of a word **within a specific context**. It involves mapping words to their **appropriate sense** in a given context to ensure accurate interpretation and understanding of text

Word sense disambiguation: different approaches

1. Dictionary-based, Knowledge-based Methods
2. Supervised Methods
3. Semi-supervised Methods

WSD (Approaches)

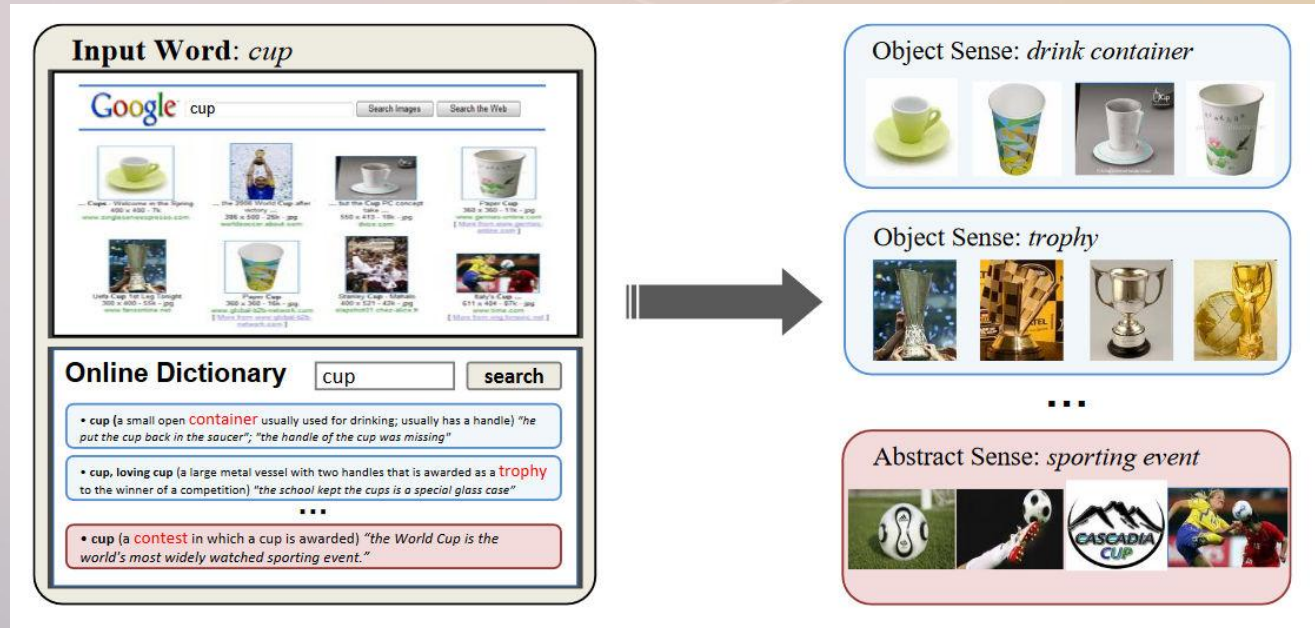
- **Dictionary-based:** measure **overlap between sense definitions** of word and current context, which further means identify the correct sense for **one word at a time**. Here the current context is the set of words in surrounding sentence or paragraph. In linguistics, **gloss definition of a word** = meaning of word in a text
- **Supervised Methods:** The context is represented as a set of features of the words. It includes the information about the surrounding words also. These methods rely on substantial amount of **manually sense-tagged corpora**, which is very expensive to create
- **Semi-supervised:** Due to the lack of training corpus, most of the WSD algorithms use semi-supervised learning. The technique that is used by semi-supervised methods is **bootstrapping from seed data**

WSD (Examples)

The **bank** can guarantee **deposits** will eventually cover future tuition costs because it invests in adjustable-rate **mortgage** securities.

bank ¹	Gloss:	a financial institution that accepts deposits and channels the money into lending activities
	Examples:	“he cashed a check at the bank”, “that bank holds the mortgage on my home”
bank ²	Gloss:	sloping land (especially the slope beside a body of water)
	Examples:	“they pulled the canoe up on the bank”, “he sat on the bank of the river and watched the currents”

Word	SID	Sense Definition	N
HARD	HARD ₁	difficulty	3,455
	HARD ₂	laborious, heavy	502
	HARD ₃	contradiction to soft	376



Source: <https://devopedia.org/word-sense-disambiguation>

WSD (Applications)

Machine Translation: Ensuring accurate translation by understanding the correct sense of words.

Information Retrieval: Helping users find more relevant information by understanding the intended meaning of search queries.

Question Answering: Enabling computers to understand the meaning of questions and provide accurate answers.

Text Summarization: Improving the quality of summaries by understanding the key meanings of words.

Sentiment Analysis: Providing a more nuanced understanding of text sentiment by identifying the correct sense of words.

WSD (Limitations)

Ambiguity of Word Meanings:

- Many words have multiple senses, and determining the correct sense in a given context is difficult..

Complexity of Context:

- Determining the correct sense of a word often requires considering the surrounding context.

Differences between Dictionaries and Thesauruses:

- Different dictionaries and thesauruses may define word senses differently, leading to inconsistencies

Availability of Sense-Annotated Corpora:

- Supervised WSD methods rely on large, high-quality, sense-annotated corpora (labeled datasets)

Word sense disambiguation (WSD)

Dictionary-based approach to WSD

In []:

```
import nltk
from nltk.wsd import lesk
from nltk.tokenize import word_tokenize
nltk.download('wordnet')
nltk.download('punkt')
nltk.download('omw-1.4')

# Example sentence
sentences = ["I went to the bank to deposit money.", "The river bank was full of mud."]
# Target word
target_word = "bank"

for sentence in sentences:
    # Tokenize the sentence
    tokens = word_tokenize(sentence)

    # Apply Lesk algorithm for WSD
    sense = lesk(tokens, target_word)

    # Output the best sense
    print(f"Sentence: '{sentence}'")
    print(f"Target word: '{target_word}'")
    print(f"Best sense: {sense}")
    print(f"Definition: {sense.definition()}")
    print()
```

Use of the Lesk algorithm

```
Out[]: Sentence: 'I went to the bank to deposit money.'
Target word: 'bank'
Best sense: Synset('savings_bank.n.02')
Definition: a container (usually with a slot in the top) for keeping money at home

Sentence: 'The river bank was full of mud.'
Target word: 'bank'
Best sense: Synset('deposit.v.02')
Definition: put into a bank account
```